

$$\begin{aligned}
&= (A \cdot (C \cdot 0) + (B'A) \cdot (A'D)) + ((BC') \cdot (CD') + C \cdot (D' \cdot (DA'))) \quad (\text{Postulate 3(b)}) \\
&= (A \cdot 0 + (B'A) \cdot (A'D)) + ((BC') \cdot (CD') + C \cdot (D' \cdot (DA'))) \quad (\text{Theorem 2(b)}) \\
&= (0 + (B'A) \cdot (A'D)) + ((BC') \cdot (CD') + C \cdot (D' \cdot (DA'))) \quad (\text{Theorem 2(b)}) \\
&= (0 + B' \cdot (A \cdot (A'D))) + (B \cdot (C' \cdot (CD')) + C \cdot (D' \cdot (DA'))) \quad (\text{Theorem 4(b)}) \\
&= (0 + B' \cdot ((A \cdot A') \cdot D)) + (B \cdot ((C' \cdot C) \cdot D') + C \cdot ((D' \cdot D) \cdot A')) \quad (\text{Theorem 4(b)}) \\
&= (0 + B' \cdot ((A \cdot A') \cdot D)) + (B \cdot ((C' \cdot C) \cdot D') + C \cdot ((D' \cdot D) \cdot A')) \quad (\text{Postulate 3(b)}) \\
&= (0 + B' \cdot (0 \cdot D)) + (B \cdot (0 \cdot D') + C \cdot (0 \cdot A')) \quad (\text{Postulate 5(b)}) \\
&= (0 + B' \cdot (0 \cdot 0)) + (B \cdot (0' \cdot 0) + C \cdot (A' \cdot 0)) \quad (\text{Postulate 3(b)}) \\
&= (0 + B' \cdot 0) + (B \cdot 0 + C \cdot 0) \quad (\text{Theorem 2(b)}) \\
&= (0 + 0) + (0 + 0) \quad (\text{Theorem 2(b)}) \\
&= 0 + 0 \quad (\text{Postulate 2(a)}) = 0 \quad (\text{Postulate 2(a)})
\end{aligned}$$

The complement of 0 is 1.

$$\begin{aligned}
(b) & B'D + A'B'C' + ACD + A'BC \\
&= B'D + (A'B'C' + ACD + A'BC) \quad (\text{Theorem 4(a)}) \\
&= B'D + (A'BC + A'BC' + ACD) \quad (\text{Postulate 3(a)}) \\
&= B'D + (A'BC + A'BC') + ACD \quad (\text{Theorem 4(a)}) \\
&= B'D + (A'B(C+C')) + ACD \quad (\text{Postulate 4(a)}) \\
&= B'D + (A'B \cdot 1 + ACD) \quad (\text{Postulate 5(a)}) \\
&= B'D + (A'B + ACD) \quad (\text{Postulate 2(b)}) \\
&= DB' + (A'B + DAC) \quad (\text{Postulate 3(b)}) \\
&= D(A'B + DAC) + DB' \quad (\text{Postulate 3(a)}) \\
&= A'B + (DAC + DB') \quad (\text{Theorem 4(a)}) \\
&= A'B + (D \cdot (AC + DB')) \quad (\text{Theorem 4(b)}) \\
&= A'B + D \cdot (AC + B') \quad (\text{Postulate 4(a)}) \\
&\rightarrow (A'B + (DAC + DB'))' = (A'B)' \cdot (DAC + DB')' \quad (\text{Theorem 5(a)}) \\
&= (A' + B)' \cdot (DAC + DB')' \quad (\text{Theorem 5(b)}) \\
&= (A + B)' \cdot (DAC + DB')' \quad (\text{Theorem 3}) \\
&= (A + B)' \cdot (DAC)' \cdot (DB')' \quad (\text{Theorem 5(a)}) \\
&= (A + B)' \cdot (D' + A' + C') \cdot (D' + B')' \quad (\text{Theorem 5(b)}) \\
&= (A + B)' \cdot (D' + A' + C') \cdot (D + B) \quad (\text{Theorem 3})
\end{aligned}$$